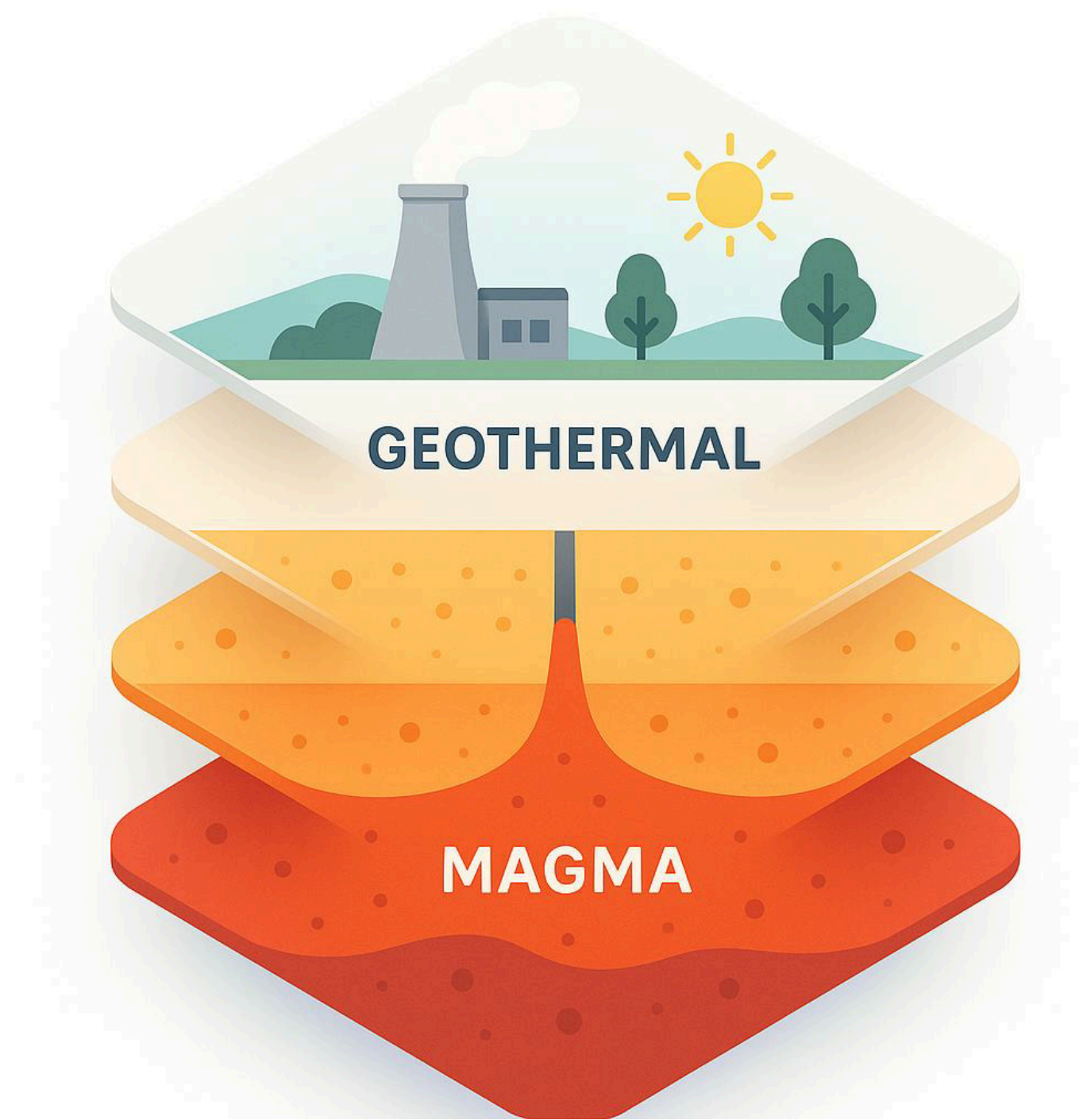


GEODE Datathon

Team: Operation Hotspot

Emmanuel Ikpesu
Fehmi Ozbayrak

The University of Texas at Austin



Engineering Challenge

Agenda

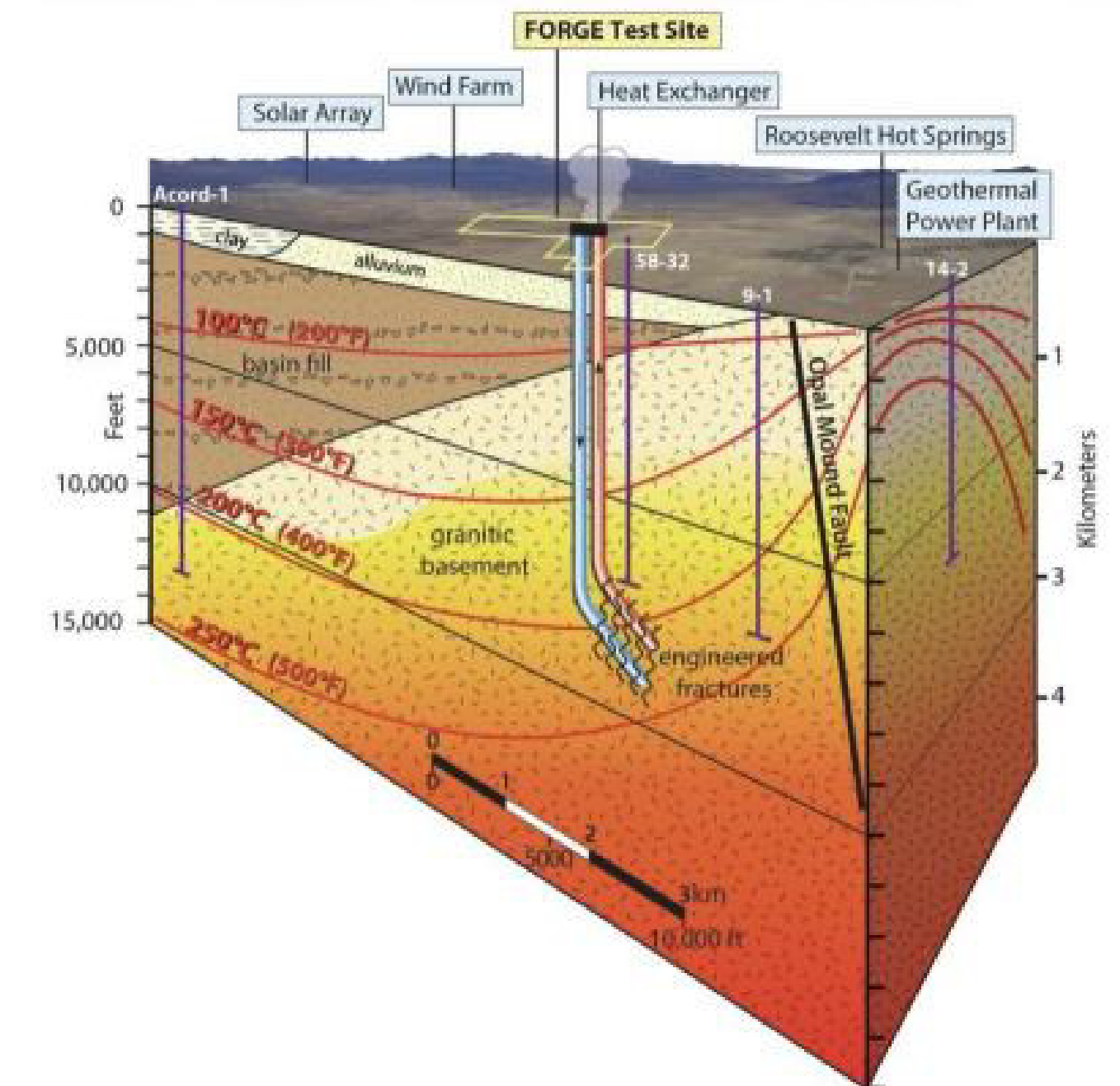
The Challenge	Part 1
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The Challenge

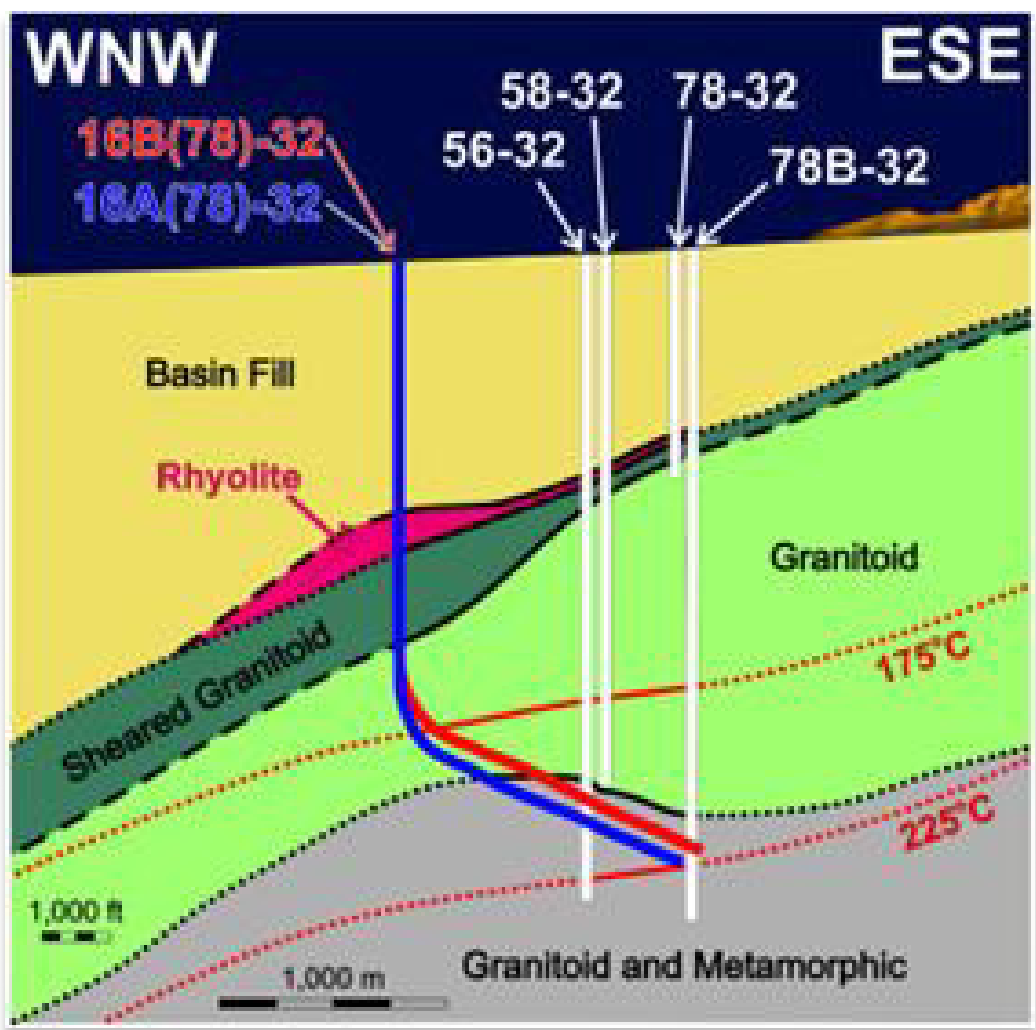
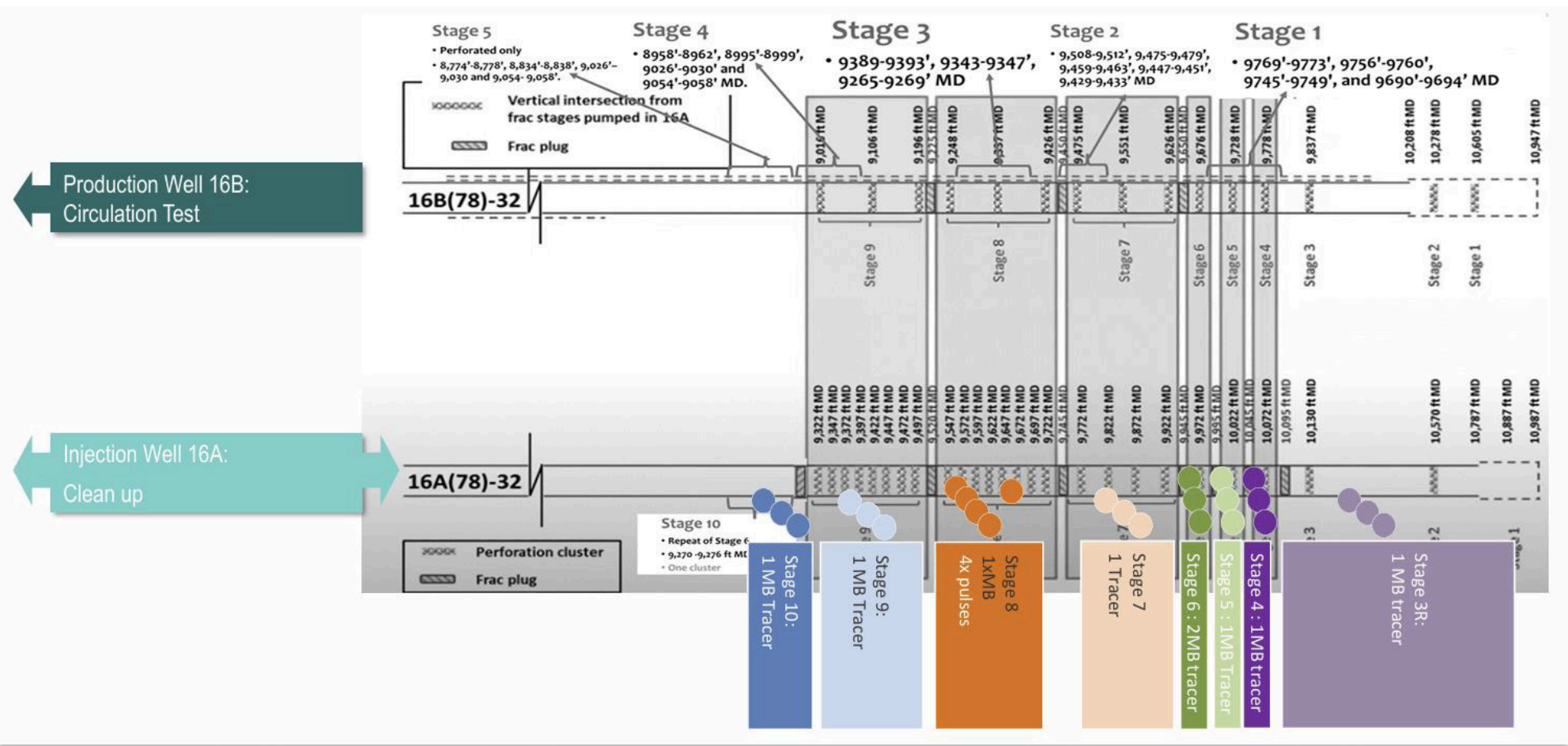
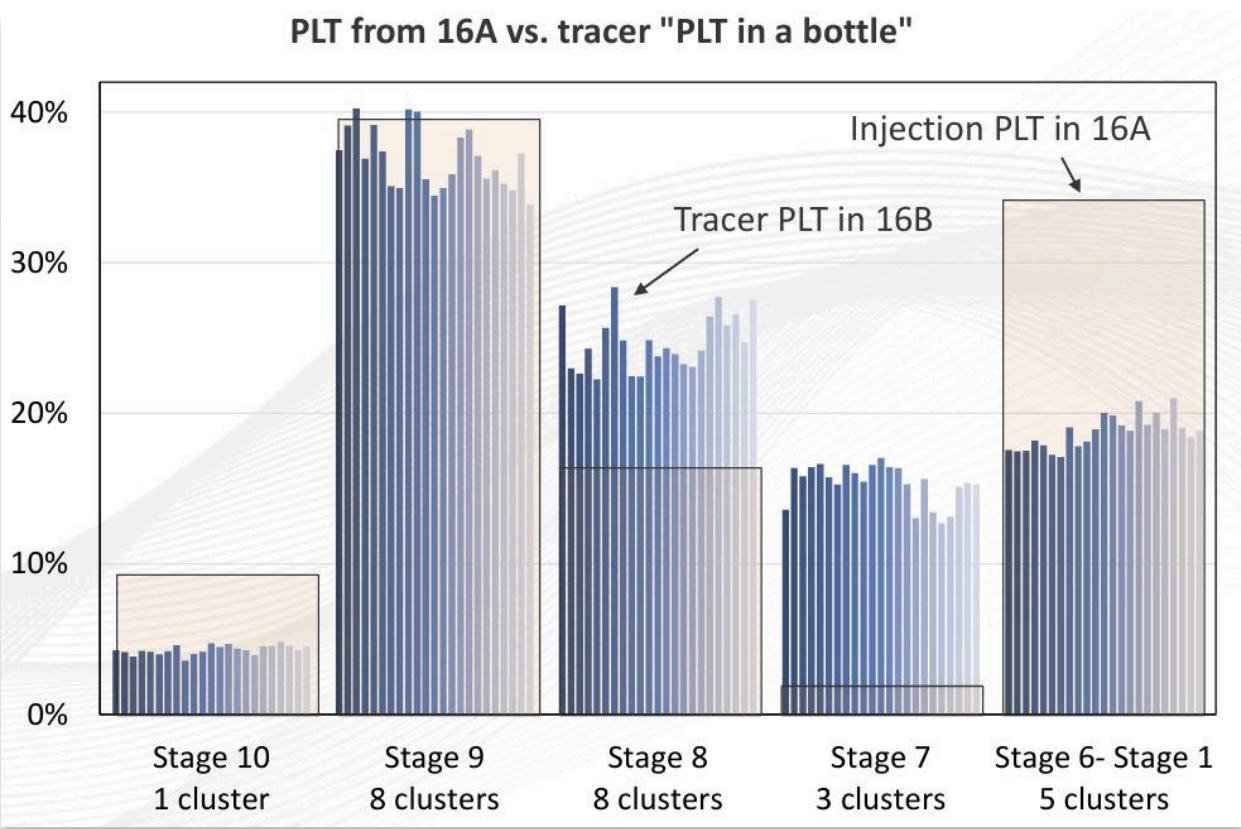
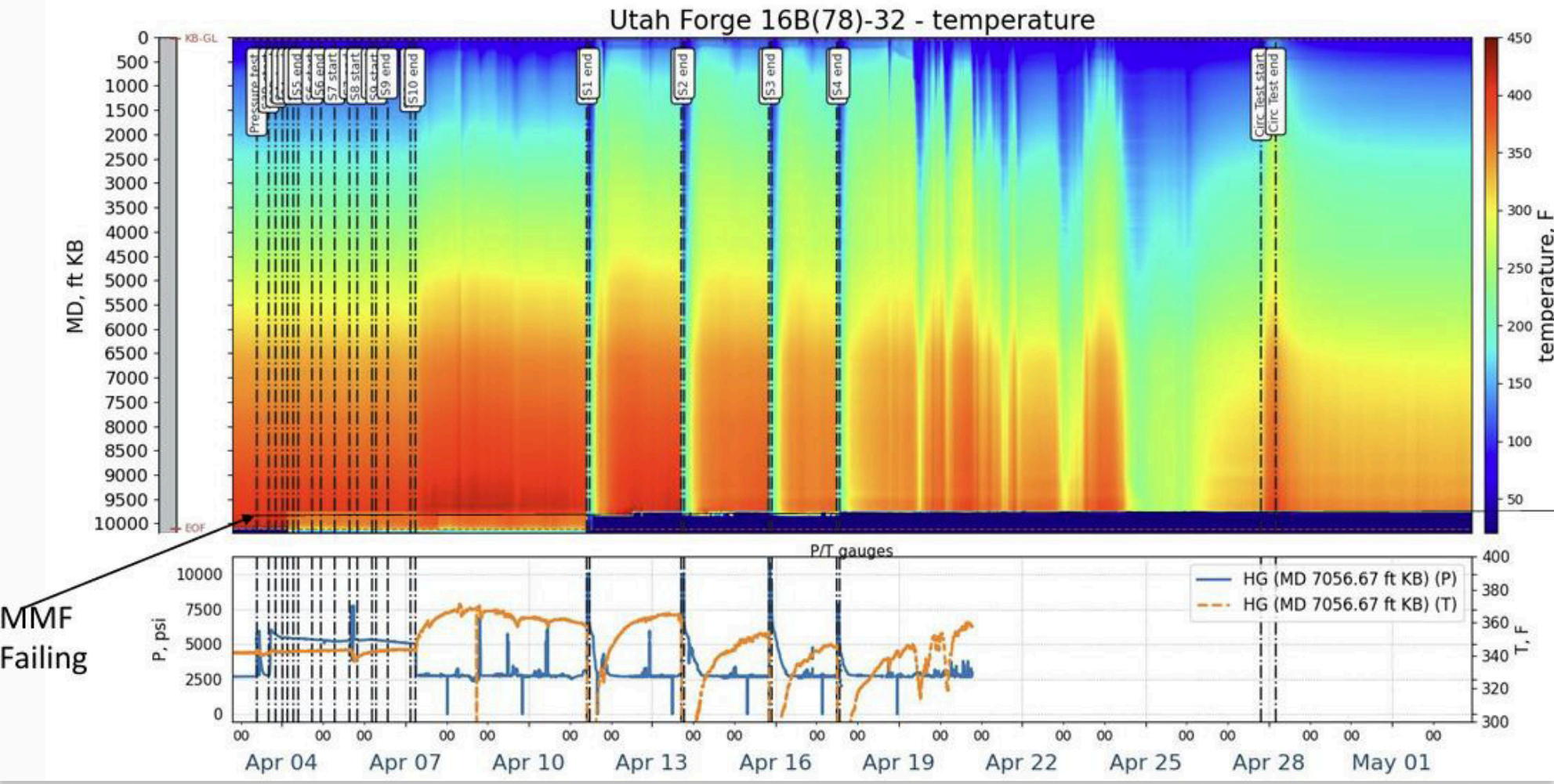
Utah FORGE: Demonstrating Enhanced Geothermal System (EGS) feasibility via hydraulic fracturing in granite.

Challenge: Leverage multi-source datasets to develop a data-driven framework for optimizing geothermal energy production

Our Goal: Utilize particular datasets, **perform investigations and data analytics** to identify insights for **enhancing EGS performance**



Investigation – April Stimulation



A comprehensive set of operations (4-28 April 2024)

- 16A Stimulation
- 16B Stimulation
- Short circulation test

A broad array of datasets

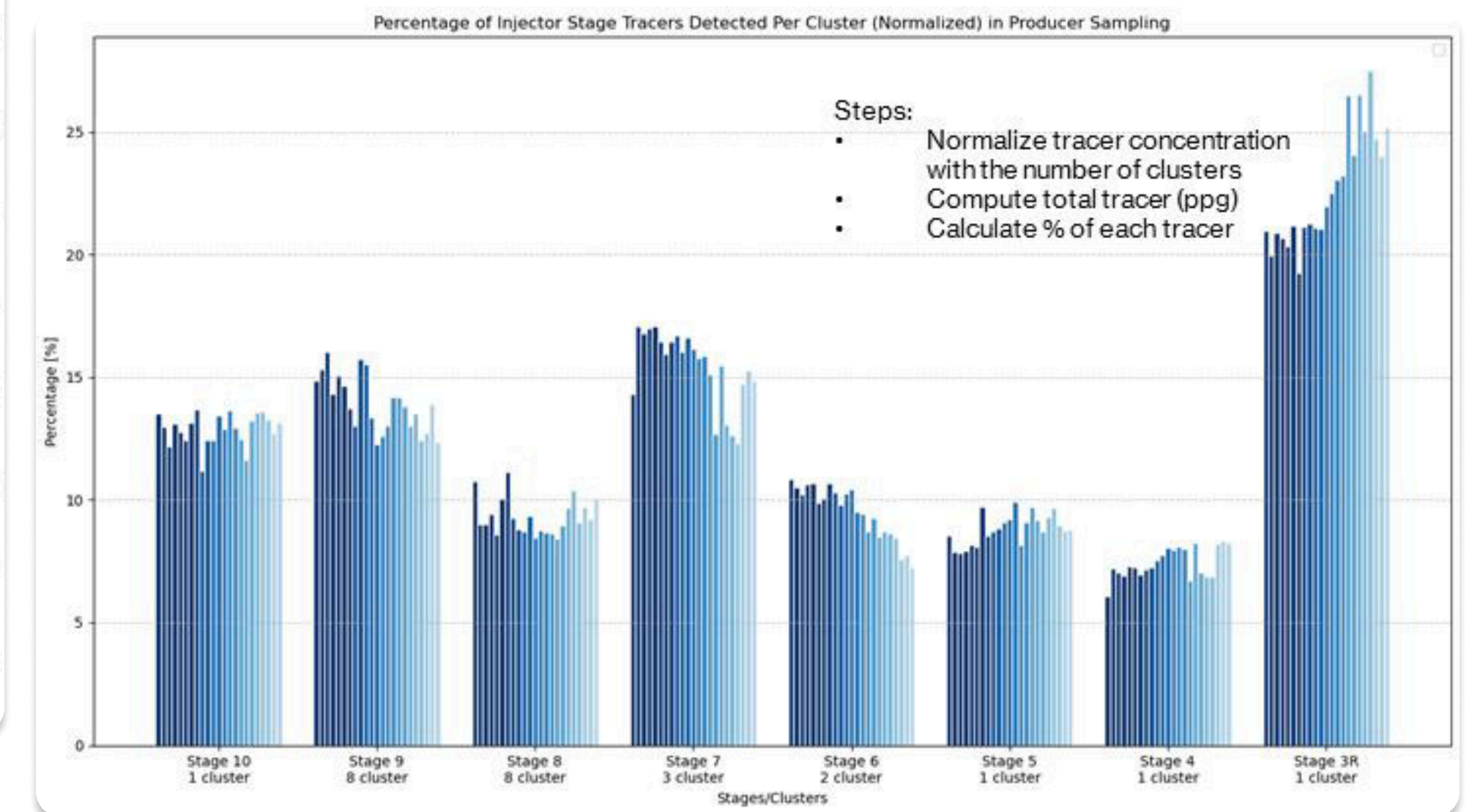
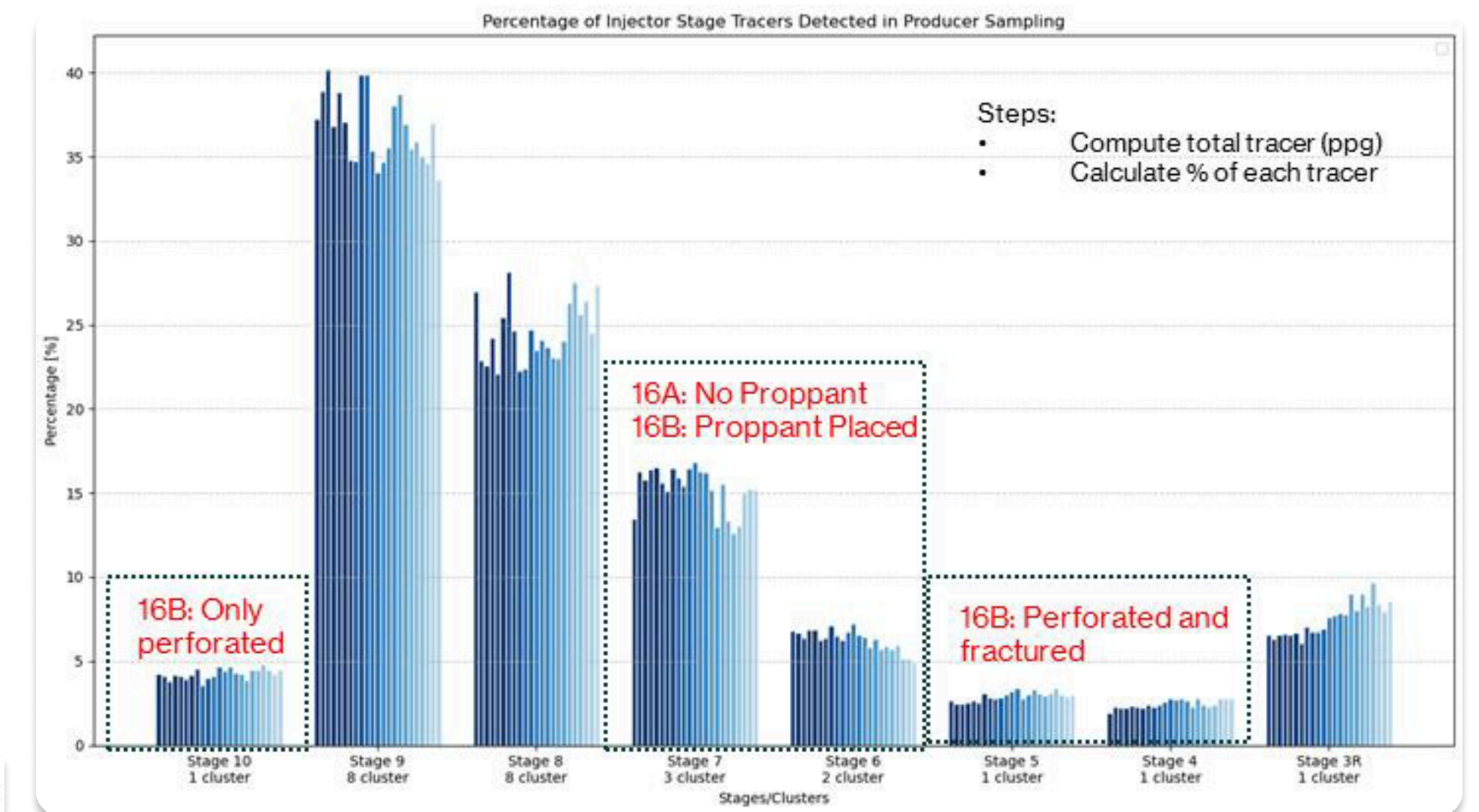
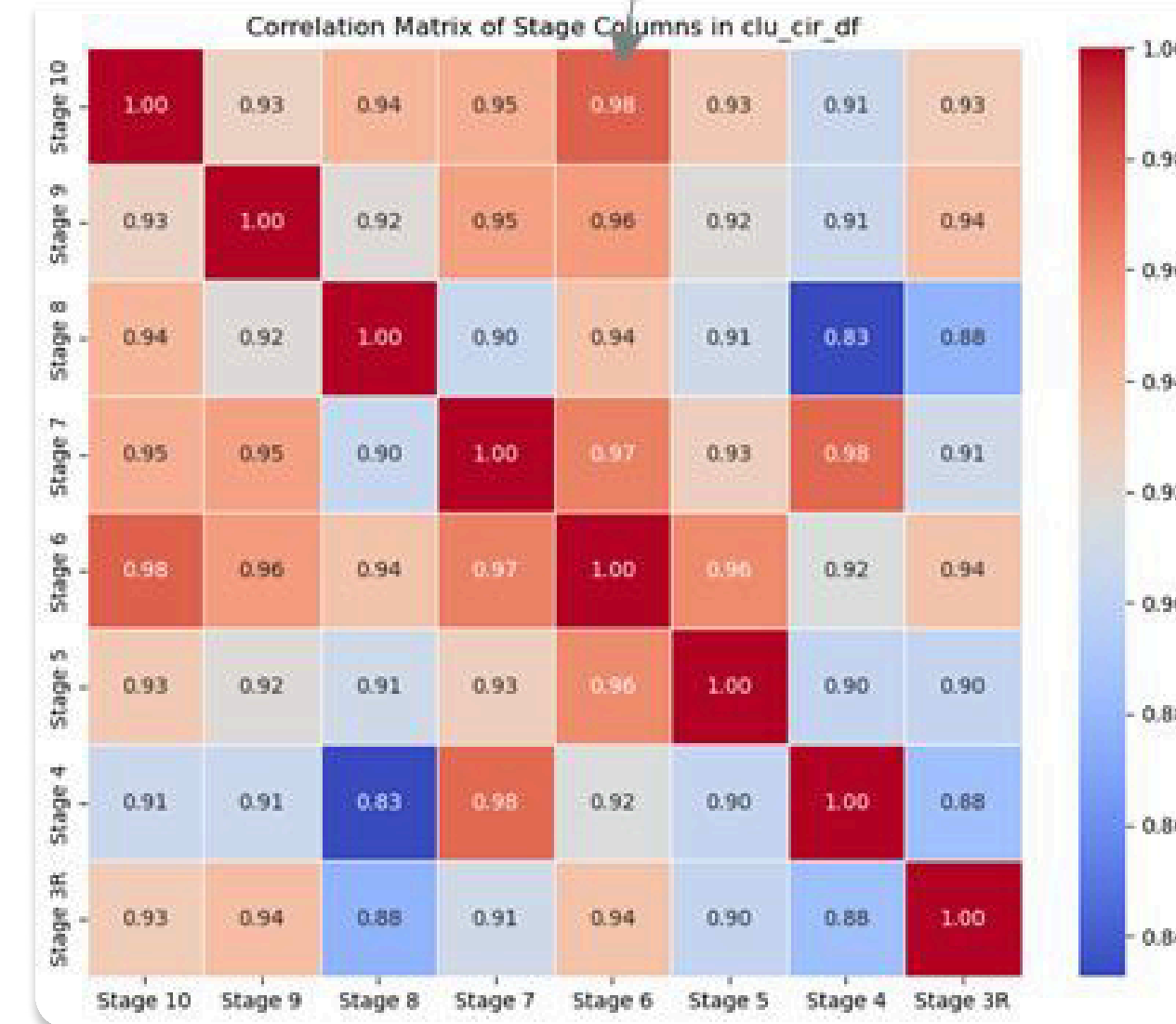
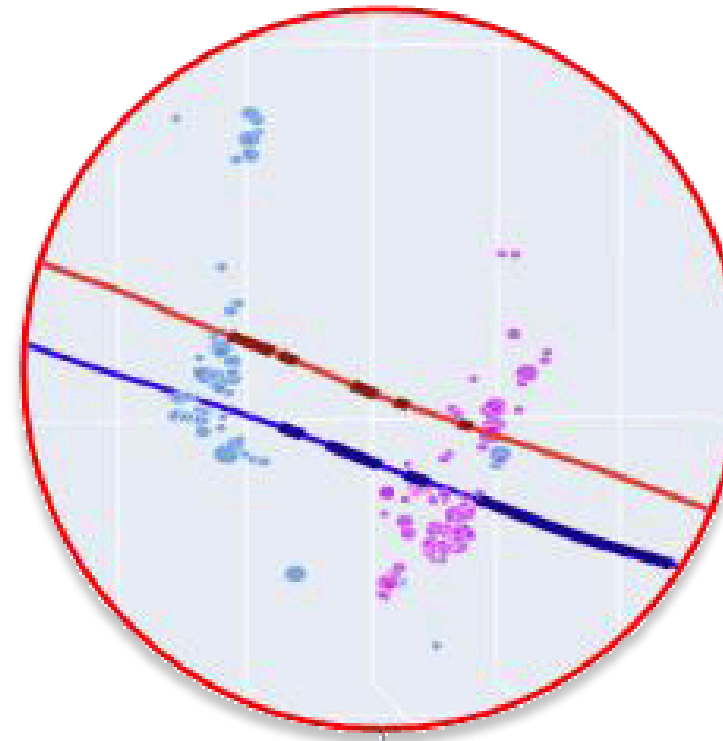
- Fiberoptic (DAS, DSS, DTS)
- Tracer test
- P&T gauges
- Microseismic Events
- Detailed wellbore surveys
- Stimulation data with stage information
- We reviewed the existing reports to gain insights.
- We preprocessed some of the data to have further insights.**

Some of the sample visuals of the datasets that we have used in our workflow. Fiberoptic data (top left), tracer data (top right), stimulation data with stage information (bottom left) and wellbore surveys (bottom right). Retrieved from <https://gdr.openet.org/>

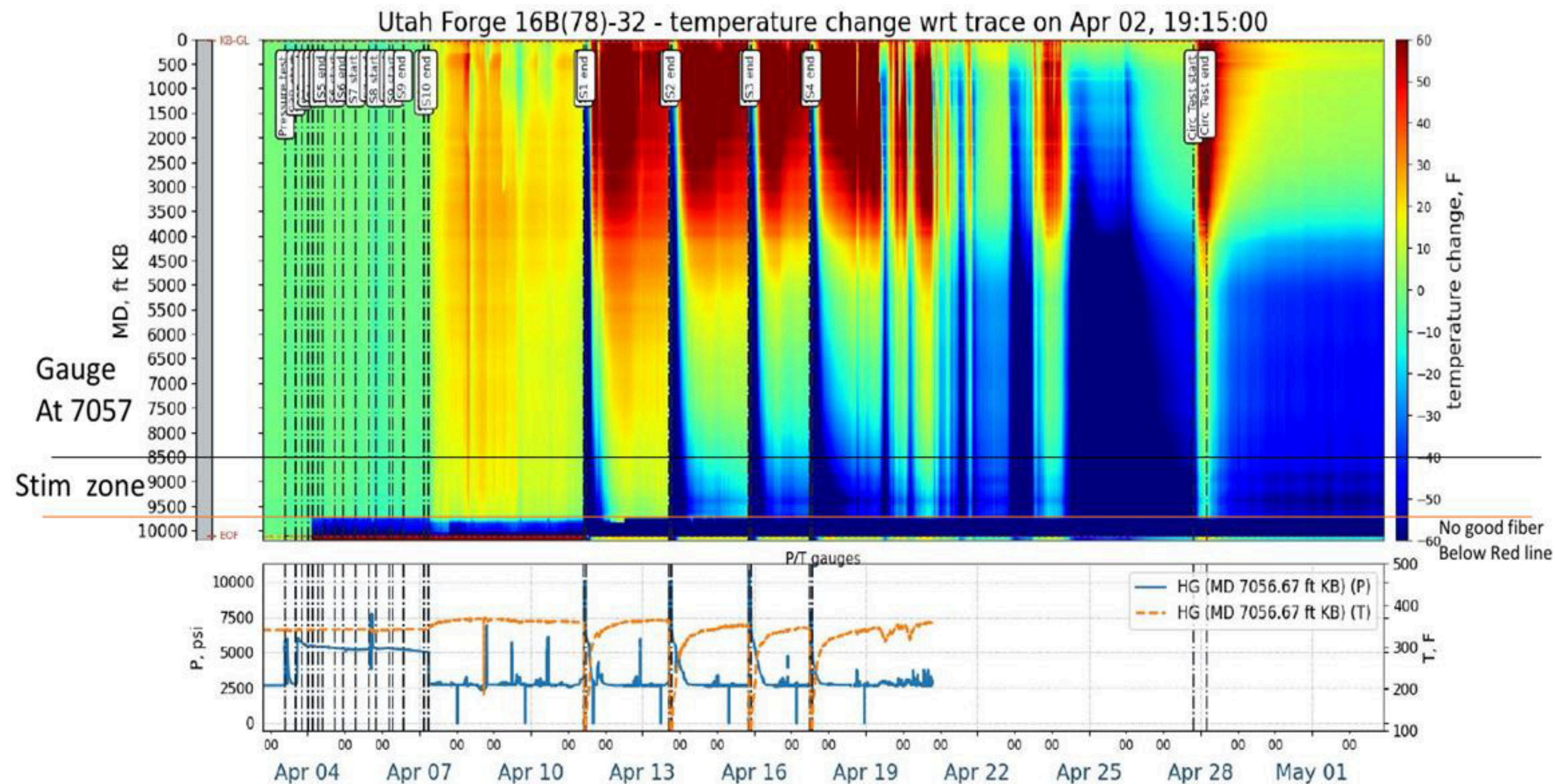
Tracer Test Insights

During the circulation test

- **No. of clusters per stage** significantly **impacts connectivity**, with more clusters improving it.
- When normalized, **each stage exhibits approximately equal connectivity**, indicating a comparable normalized effectiveness for each cluster.
- The correlation plot of recovered tracer concentrations shows **high correlation** for the majority of stages. This implies that flow paths from these stages are interconnected or share similar hydraulic properties. However, Stage 4 notably displays low correlation.



Fiberoptic Data Insights



DTSDelta T plot for April Stimulation and Circulation Test. Retrieved from <https://gdr.openei.org>. Courtesy of Neubrex.

16B Stimulation and Circulation Test

- **Preferential flow path** observed after **S2**. It continues until the end of stimulation, and it is **apparent in circulation test**. Path depth corresponds to S2 as well.
- Same phenomenon observed after **S4**. Path depth corresponds to S4 as well.
- Preferential flow paths are either **created or enhanced** during S2 and S4.
- S1 might have the same situation, but fiber fails at S1 depth.

Data Analytics – Microseismic Events

Nature of microseismic events

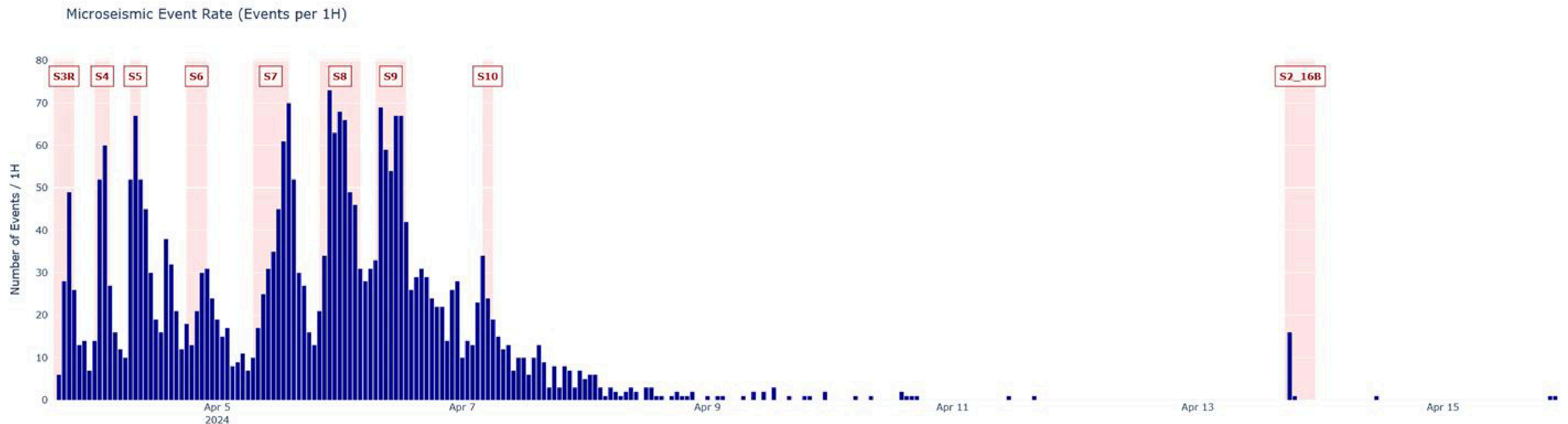
During stimulation, numerous **microseismic events** occur across **different zones.** When the **main fracture initiates and propagates,** a **sharp increase in activity** is observed.

This leads to the **highest rate of microseismic events per unit time** during the **fracture propagation phase.**

Microseismic Events Per Unit Time

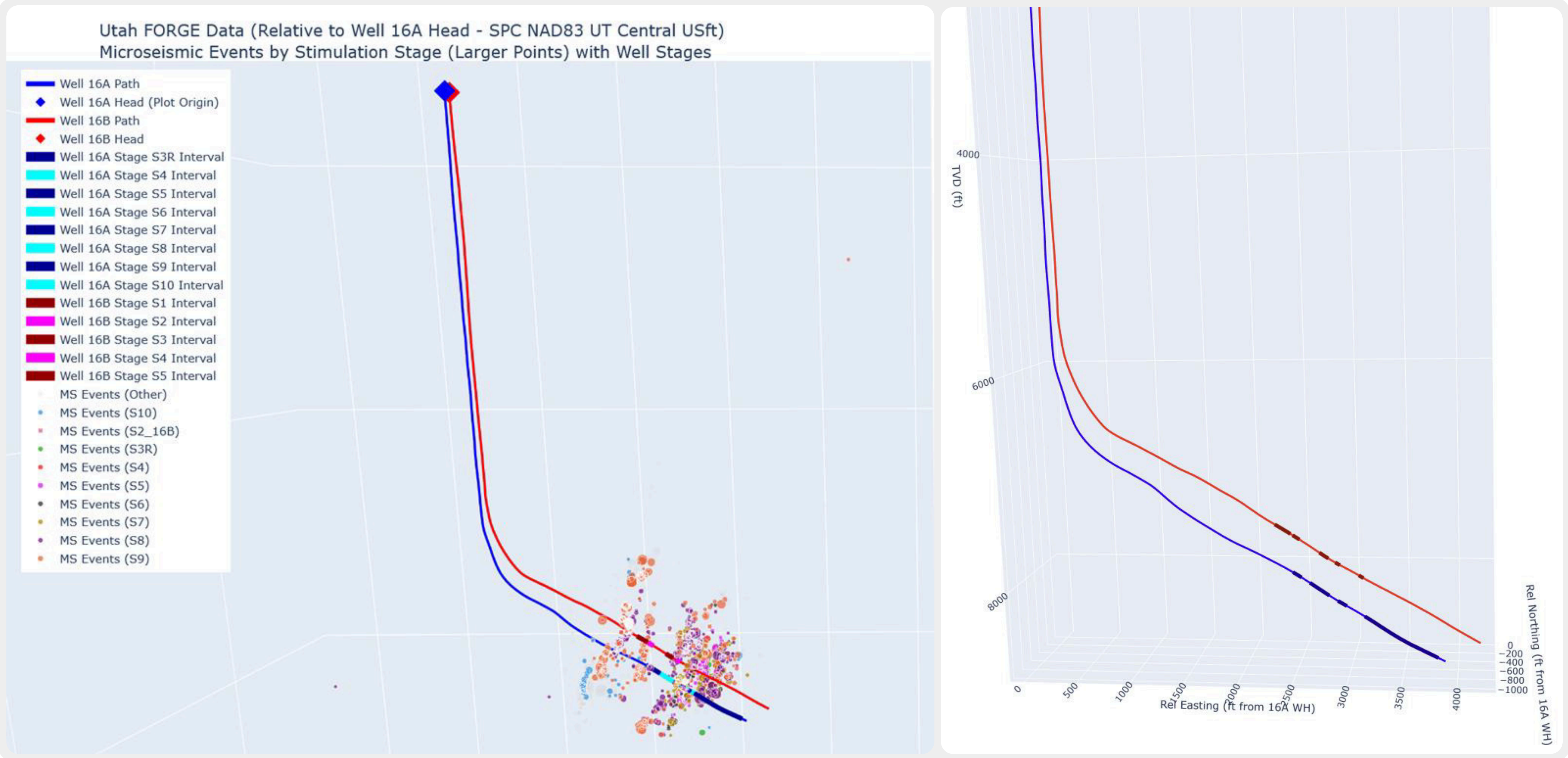
Identification of the "Peak Events" reveals critical information about generated fractures

- As expected, these peaks **corresponded to stimulation stages**.
- **Lack of catalogued data after April 15** obscures important information.
- These peak events are used for further analysis.



*Microseismic event catalogue binned by 1H intervals to identify events occur at peak event rates. Retrieved from <https://gdr.openei.org>

Microseismic Events Per Unit Time



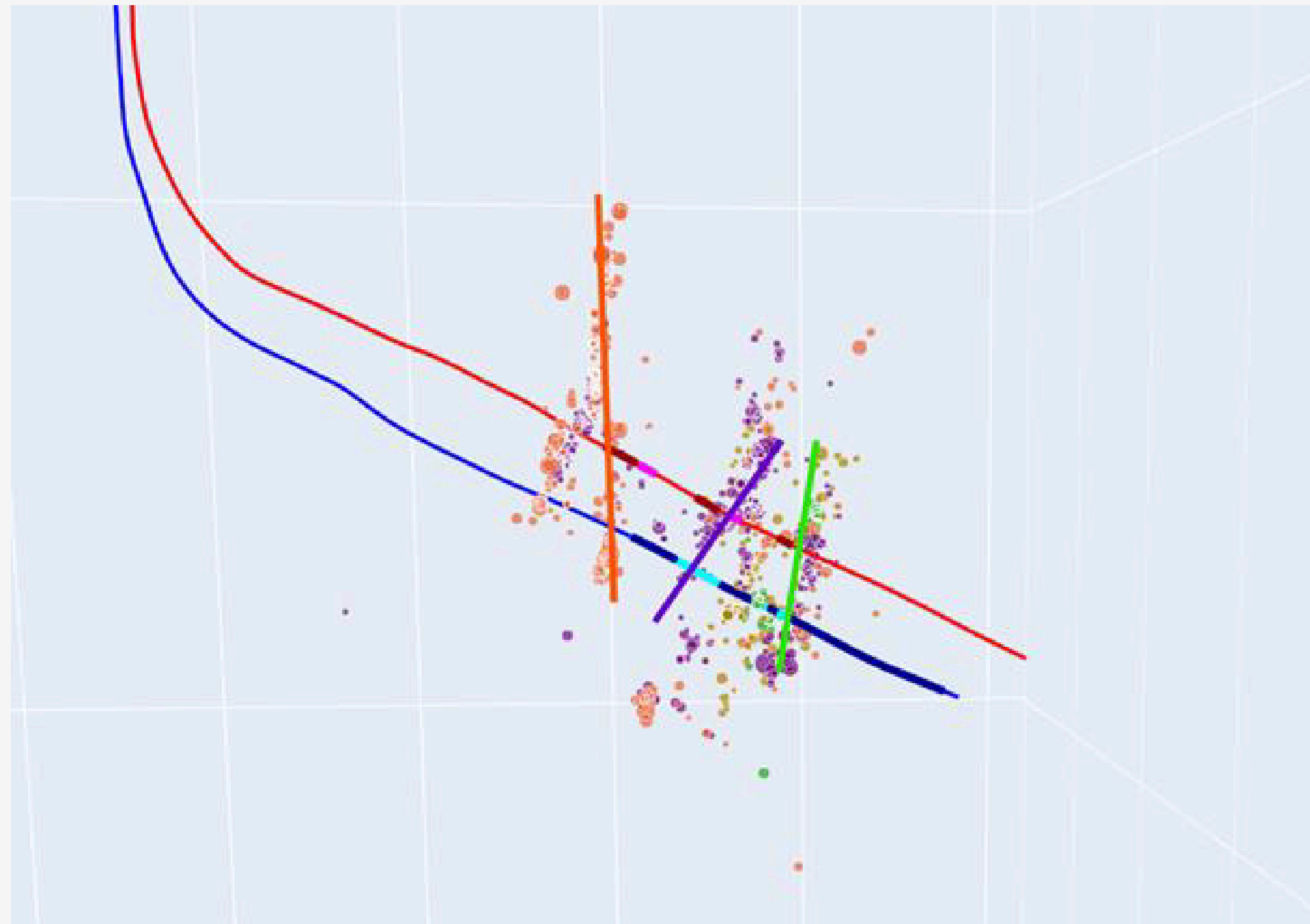
*Visualization of wellbore surveys and microseismic events occurred at peak event times. Retrieved from <https://gdr.openei.org>

*Slide deck will have static images, but the video presentation will have interactive plots. Interactive plots will be delivered along with the submission.

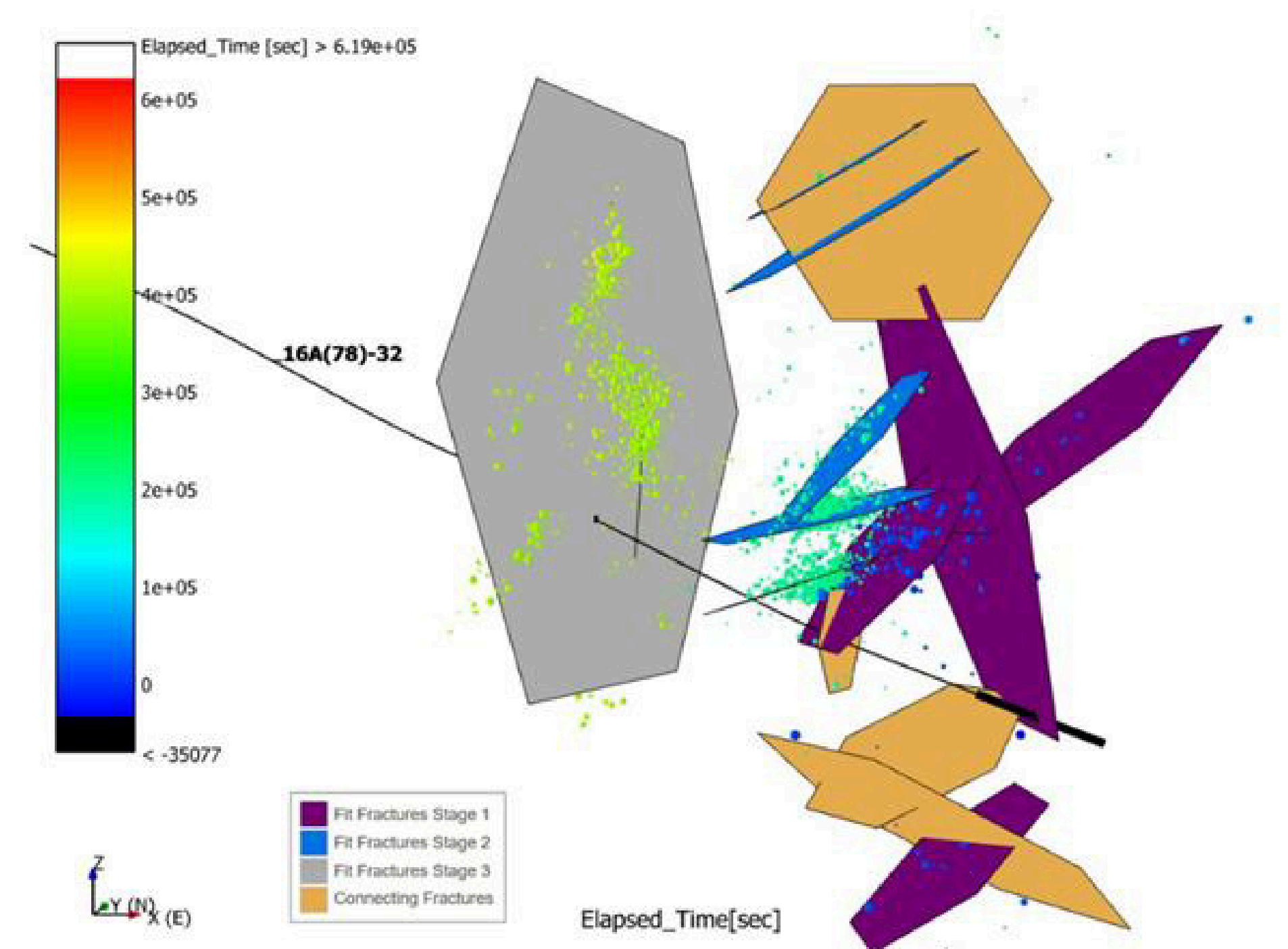
Data Analytics – Visualization

The Stress shadow effect is prominent

- Findings are aligned with Finnila et al (2023) and Finnila et al (2024).
- We believe there are **two dominant flow paths reached & enhanced by 16B S2 & S4**.
- S3R flow path is anticipated to be dominant as well, but we can't confirm due to lack of DTS data.



*Visualization of wellbore surveys and microseismic events occurred at peak event times. Retrieved from <https://gdr.openet.org>



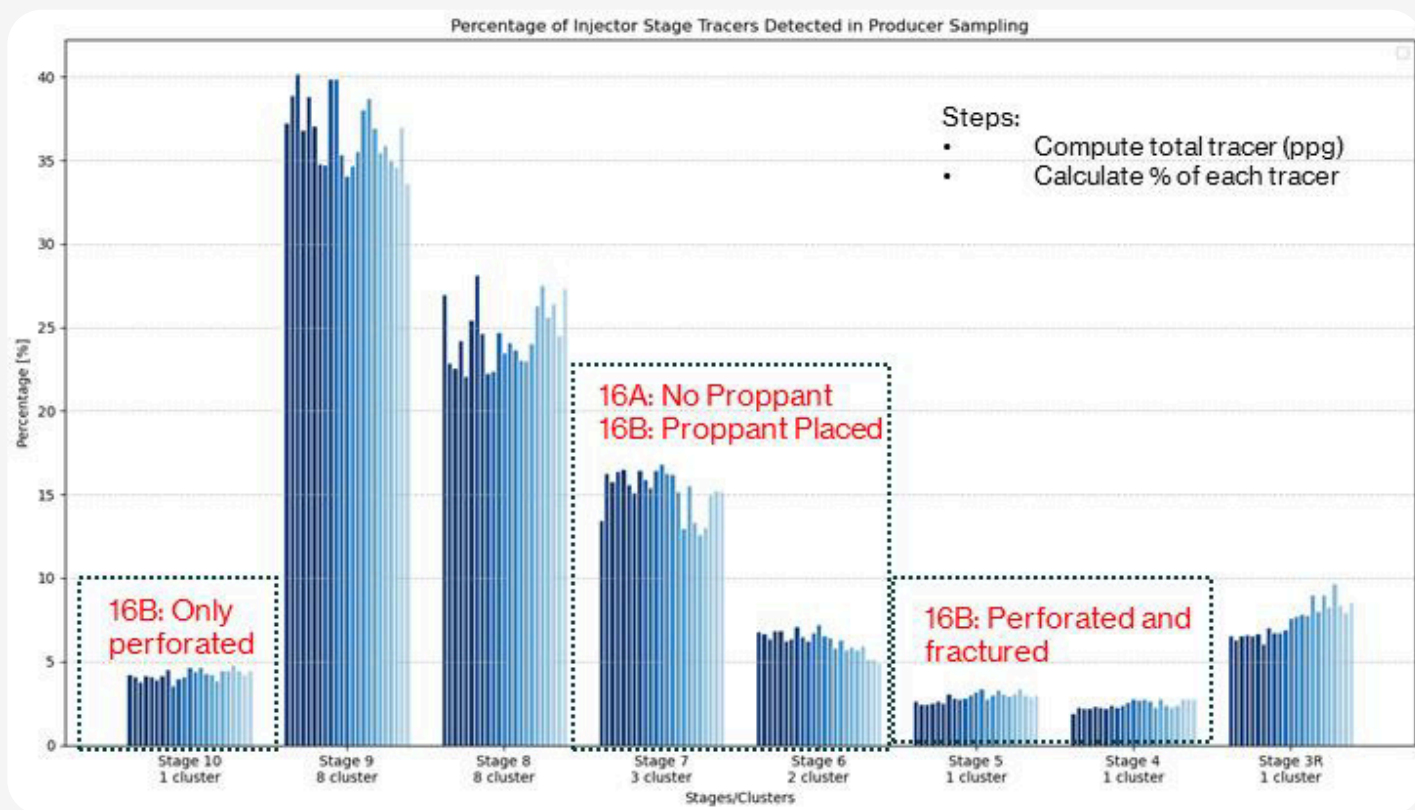
Proposed fracture network by Finnila et al (2023). 3 main fractures (2, X shaped purple fractures and gray fracture) are also obtained from our analysis, and found to be dominating the network. Notice how the blue fractures are also "going" towards the purple fractures, indicates the "sink" effect of existing fractures, indicating stress shadow. Retrieved from Finnila et al (2023)

Results

1

Fractures for all stages are successfully generated

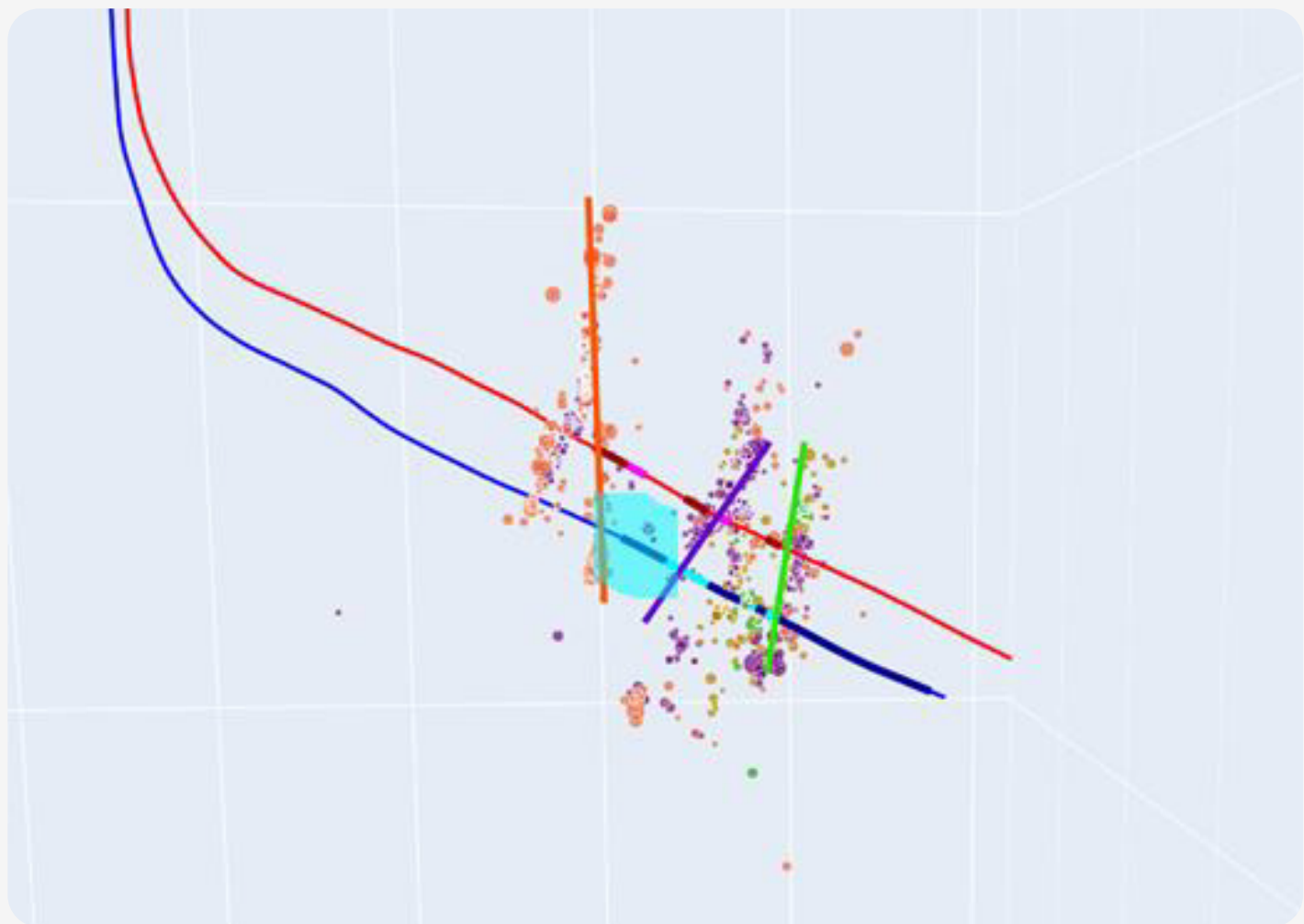
- Tracer test shows **connection between 16A and 16B**.
- A **strong relationship** is observed between the **contribution and the number of clusters per stage**.



2

Dominating flow paths act as "sink" – stress shadow

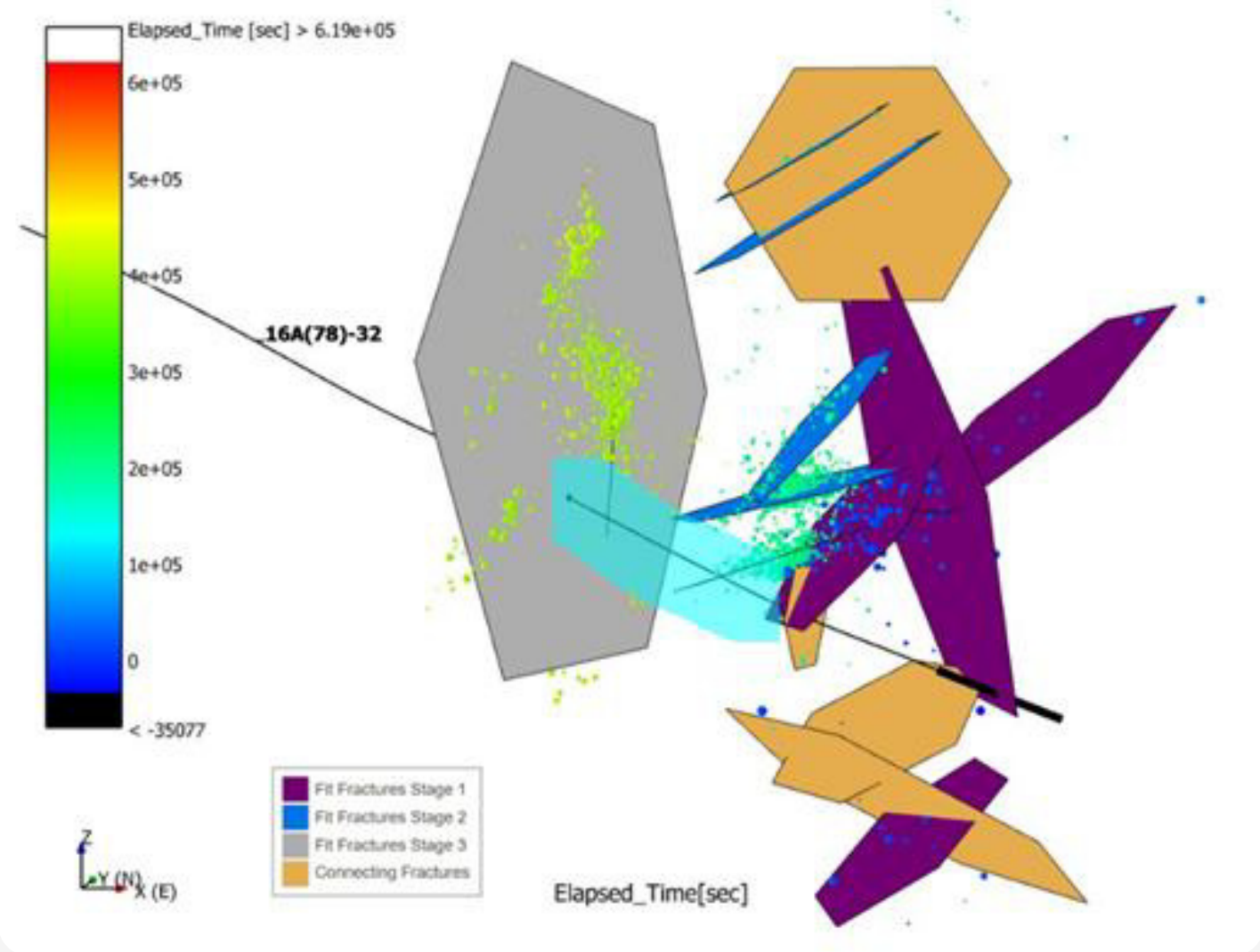
- Observed both in **DTS and MS** event catalogue.
- This may cause **thermal short-circuiting** in the long run.



3

Under-utilized reservoir volume located between the dominant flow paths

- A careful **re-stimulation** between these fractures may increase the flow uniformity, EGS efficiency and thermal breakthrough lifetime.



Discussion – What could be useful



Low-hanging fruit: Microseismic events catalogued for 16B stimulation

- Better understanding of enhanced flow paths & directions.



We suggest a Flow Scanner Imager test

- Opportunity to identify dominant flow paths and investigate the validity of our findings.
- Will provide essential information about fracture associated with S3R – missing fiber data obscures a detailed analysis of this fracture.
- Spinner log is a budget alternative.

Suggestion for Restimulation

- ✓ **A detailed perturbed stress analysis**
 - **Quantification of the altered stress field** is critical for optimizing the re-stimulation design.
- ✓ **Perforation design**
 - **Oriented perforating techniques**, with the azimuth guided by the stress perturbation analysis.
 - **XLET** across perforation clusters within the new stage to ensure uniform treatment.
- ✓ **Mechanical Isolation of the Target Interval**
 - Utilizing **straddle packers** is a robust option.
 - Alternatively, or as a complement, degradable particulate **diverters** could be considered.
- ✓ **High Viscosity Pulsed Injection for initiation**
 - Employ a high-viscosity fluid pill for fracture initiation, potentially combined with pulsed injection.
- ✓ **Regular injection after fracture has been initiated**
 - Following successful initiation with the specialized pill, transition to the main EGS stimulation fluid.

Thank You!

References



Finnila, A., Damjanac, B., & Podgorney, R. (2023, February).

Development of a Discrete Fracture Network Model for Utah FORGE using Microseismic Data Collected During Stimulation of Well 16A (78)-32. In 48th Workshop on Geothermal Reservoir Engineering, Stanford University, Stanford, California.



Finnila, A., & Jones, C. (2024). Utah FORGE

2024 Discrete Fracture Network Model Data (No. 1646). USDOE Geothermal Data Repository (United States); Energy and Geoscience Institute at the University of Utah.